

BI Oral Questions

1. What is Business Intelligence (BI)? How is it different from traditional data analysis?

Business Intelligence (BI) refers to the technologies, processes, and tools that transform raw data into meaningful insights for decision-making. It includes reporting, analytics, dashboards, and data visualization.

Difference: Traditional data analysis is often manual and reactive, involving spreadsheets or basic queries, whereas BI is more automated, scalable, and interactive, enabling real-time, data-driven decisions.

2. What are the major components of a BI architecture?

- **Data Sources** (databases, files, APIs)
 - **ETL Tools** (Extract, Transform, Load processes)
 - **Data Warehouse/Data Marts**
 - **OLAP Cubes**
 - **BI Tools** (e.g., Power BI, Tableau)
 - **Dashboards and Reports**
 - **Metadata & Master Data Management**
 - **Data Governance and Security**
-

3. Can you explain the BI lifecycle?

The BI lifecycle typically includes:

1. **Data Collection** – Gathering data from multiple sources
 2. **Data Integration** – Using ETL to clean and load data
 3. **Data Storage** – Saving structured data in warehouses or marts
 4. **Data Analysis** – Applying data mining, OLAP, or statistical tools
 5. **Reporting and Visualization** – Dashboards, KPIs, and interactive reports
 6. **Decision Making** – Using insights to drive business actions
 7. **Monitoring and Feedback** – Continuous improvement based on usage and outcomes
-

4. What is the role of a data warehouse in BI?

A data warehouse is a centralized repository that stores integrated data from multiple sources. It supports analytical reporting, structured queries, and decision-making by providing historical, consistent, and clean data.

5. How do OLAP and OLTP systems differ?

PDF BY VL



Telegram Channel

[Click here to join](#)



Instagram Page

[Click here to join](#)



WhatsApp Channel

[Click here to join](#)



WHY JOIN ?

- 🏆 Trusted by 80,000+ Engineering Students
- 🔔 Stay Updated with latest university news
- 🚀 Make your engineering journey easy with us

Best Regards, **Vishwajeet Londhe**

https://t.me/SPPU_TE_BE_COMP

© 2020 Vishwajeet Londhe. All Rights Reserved

- **OLAP (Online Analytical Processing):** Used for complex queries and analysis (BI); optimized for reads.
 - **OLTP (Online Transaction Processing):** Used for daily operations like order entry, payments; optimized for fast transactions and data writes.
-

6. What is ETL and why is it important?

ETL stands for **Extract, Transform, Load**. It is the process of:

- **Extracting** data from source systems
- **Transforming** it into a consistent format
- **Loading** it into a data warehouse

ETL is critical because it ensures data quality, consistency, and readiness for analysis.

7. Define data mining and its relevance in BI.

Data mining is the process of discovering patterns, trends, and relationships in large datasets using statistical, machine learning, or AI techniques. It enhances BI by providing predictive insights and deeper understanding of data.

8. What are KPIs and how are they used in BI dashboards?

Key Performance Indicators (KPIs) are measurable values that reflect how effectively a company is achieving key objectives. In BI dashboards, KPIs are visual metrics (e.g., revenue growth, customer churn) used to monitor performance in real-time.

9. What is a data mart and how does it differ from a data warehouse?

A **data mart** is a subset of a data warehouse, focused on a specific business area (e.g., sales, finance).

Differences:

- **Scope:** Data mart is subject-specific; warehouse is enterprise-wide.
 - **Size:** Data marts are smaller.
 - **Complexity:** Data marts are simpler and faster to implement.
-

10. Explain the star schema and snowflake schema.

- **Star Schema:** A central fact table linked to dimension tables. Simpler and faster for querying.
 - **Snowflake Schema:** Dimension tables are normalized into sub-dimensions. More complex but uses less space and maintains data integrity.
-

11. Which BI tools are you familiar with (e.g., Power BI, Tableau, QlikView)?

I'm familiar with several Business Intelligence (BI) tools, including:

- **Power BI** – for interactive dashboards, DAX calculations, and seamless integration with Microsoft products.
 - **Tableau** – known for its powerful visualizations and drag-and-drop interface.
 - **QlikView/Qlik Sense** – used for associative data modeling and fast in-memory data processing.
 - Others include **Looker**, **Google Data Studio**, and **SAP BusinessObjects** depending on organizational use.
-

12. What is the role of SQL in Business Intelligence?

SQL (Structured Query Language) plays a crucial role in BI by:

- **Extracting data** from relational databases.
 - **Transforming data** (e.g., joins, aggregations) into analysis-ready formats.
 - Creating **views and stored procedures** to support reporting.
 - Being the foundational language for data modeling, used in tools like Power BI and Tableau when preparing data sources.
-

13. How does Power BI connect to different data sources?

Power BI connects to data sources in several ways:

- **Direct Connection:** Import data from files (Excel, CSV, JSON), databases (SQL Server, Oracle, MySQL), cloud services (Azure, Salesforce, SharePoint, etc.).
 - **Live Connection:** Connect to sources like SQL Server Analysis Services (SSAS) without importing data, allowing real-time updates.
 - **Power Query (M language):** Allows data transformation and cleansing before loading.
 - **APIs and Custom Connectors:** For connecting to niche or proprietary systems.
-

14. Can you describe the process of creating a dashboard in Tableau?

Creating a dashboard in Tableau involves these steps:

1. **Connect to Data** – Choose a data source and load it into Tableau.
2. **Prepare Data** – Perform data cleaning or join multiple data sources as needed.
3. **Create Sheets** – Build individual visualizations (charts, tables, maps) using drag-and-drop fields.
4. **Build Dashboard** – Drag sheets into the dashboard canvas, arrange them, and configure filters and actions.
5. **Add Interactivity** – Use filters, highlight actions, and parameters to make the dashboard dynamic.
6. **Publish** – Share via Tableau Server, Tableau Online, or export as PDF/image.

15. What are the differences between Power BI and Tableau?

Feature	Power BI	Tableau
Ease of Use	More beginner-friendly, especially for Excel users	Slightly steeper learning curve
Integration	Strong with Microsoft ecosystem (Excel, Azure, SharePoint)	Flexible with various data sources
Pricing	Generally more cost-effective	More expensive for enterprise deployment
Data Modeling	Uses DAX and Power Query (M) for complex data models	Uses calculated fields and Level of Detail (LOD) expressions
Visualization	Good variety, slightly limited customization	Highly customizable and advanced visual capabilities
Performance	Better for small to medium datasets	Scales well with large datasets and complex dashboards

16. What is data cleansing and why is it important?

Data cleansing (or data cleaning) is the process of identifying and correcting or removing inaccurate, incomplete, duplicate, or irrelevant data from a dataset.

Importance:

- Improves data quality and accuracy.
- Enhances decision-making by ensuring reliable data.
- Prevents errors in analysis, reporting, and downstream systems.

17. How do you handle missing or inconsistent data in BI systems?

Common methods include:

- **Missing Data:**
 - **Imputation:** Fill with mean, median, mode, or predictive values.
 - **Deletion:** Remove rows/columns with excessive missing values.
 - **Flagging:** Mark missing values for downstream handling.
- **Inconsistent Data:**
 - **Standardization:** Use uniform formats (e.g., date formats, units).
 - **Validation rules:** Apply business rules to detect and correct errors.
 - **Referential integrity checks:** Ensure relationships between tables are intact.

18. What is dimensional modeling?

Dimensional modeling is a design technique used in data warehouses to optimize for data retrieval and analytical processing. It organizes data into:

- **Fact tables:** Contain measurable, quantitative data (e.g., sales amount).
- **Dimension tables:** Contain descriptive attributes (e.g., product name, date).

This model typically follows a **star schema** or **snowflake schema** structure and supports fast querying and reporting.

19. Explain slowly changing dimensions with examples.

Slowly Changing Dimensions (SCDs) are dimensions in which attribute values change slowly over time.

Types:

- **Type 1:** Overwrites old data.
Example: Customer address updated directly.
 - **Type 2:** Stores history by creating a new record.
Example: A new row is added when a customer's address changes, with timestamps.
 - **Type 3:** Stores limited history in the same row using additional columns.
Example: Current and previous address columns.
-

20. What are fact tables and dimension tables?

- **Fact tables:**
 - Contain numeric metrics or "facts" of a business process.
 - Have foreign keys to dimension tables.
 - Examples: Sales amount, profit, quantity sold.
- **Dimension tables:**
 - Contain descriptive, textual data for context.
 - Examples: Customer name, product category, region.

These two types of tables are central to dimensional modeling and OLAP systems.

21. What are the different types of analytics (descriptive, predictive, prescriptive)?

- **Descriptive Analytics:** Answers "What happened?" by summarizing past data. Common tools include dashboards, reports, and KPIs.
 - *Example:* Monthly sales reports, website traffic metrics.
 - **Predictive Analytics:** Answers "What is likely to happen?" by using statistical models and machine learning to forecast future trends.
 - *Example:* Sales forecasting, customer churn prediction.
 - **Prescriptive Analytics:** Answers "What should we do?" by recommending actions based on predicted outcomes.
 - *Example:* Optimization models for pricing or logistics.
-

22. How can BI support decision-making in an organization?

BI supports decision-making by:

- Providing **real-time insights** through dashboards and reports.
 - Allowing **data-driven decisions** rather than gut-based guesses.
 - Identifying **trends, patterns, and anomalies**.
 - Helping **optimize operations** (e.g., inventory, staffing, marketing).
 - Supporting **strategic planning** with scenario analysis and forecasting.
-

23. What are some common types of visualizations and when do you use them?

- **Bar Chart:** Compare quantities across categories.
Use: Product sales by region.
 - **Line Chart:** Show trends over time.
Use: Monthly revenue or user growth.
 - **Pie Chart:** Show part-to-whole relationships.
Use: Market share distribution (limited use for few segments).
 - **Scatter Plot:** Show correlation between two variables.
Use: Sales vs. advertising spend.
 - **Heat Map:** Show intensity or concentration across a matrix or geographic region.
Use: Website click density, performance by area.
 - **Histogram:** Show frequency distribution.
Use: Age distribution of customers.
-

24. What is drill-down and roll-up in OLAP?

- **Drill-Down:** Moves from summary to more detailed data.
Example: From yearly sales → quarterly → monthly → daily.
- **Roll-Up:** Aggregates detailed data to a higher-level summary.
Example: From daily sales → monthly → quarterly → yearly.

These operations help analyze data at different levels of granularity.

25. How do you ensure a BI report is effective and user-friendly?

- **Clarity:** Use clear titles, labels, and legends.
 - **Relevance:** Show only meaningful KPIs for the target audience.
 - **Visualization:** Use appropriate charts for the data type.
 - **Consistency:** Maintain formatting and layout standards.
 - **Interactivity:** Allow filtering, drill-down, and customization.
 - **Performance:** Ensure fast loading and responsiveness.
 - **Feedback:** Involve users during development to align with their needs.
-

26. Example of How BI Helped a Business Improve Performance:

Walmart uses BI tools to analyze customer purchasing patterns and optimize inventory. By processing vast amounts of transaction data, it ensures products are stocked efficiently, reducing costs and improving customer satisfaction through better product availability.

27. How BI Can Be Used in the Healthcare/Retail/Finance Sector:

- **Healthcare:** BI helps hospitals track patient outcomes, identify treatment effectiveness, reduce costs, and predict patient admission rates.
 - **Retail:** BI tools analyze customer behavior, optimize supply chains, personalize marketing, and enhance sales forecasting.
 - **Finance:** BI supports fraud detection, risk analysis, regulatory compliance, and real-time decision-making in investments and lending.
-

28. Ethical Concerns in BI:

- **Privacy Violations:** Misuse of personal data or lack of informed consent.
 - **Bias in Data:** Leading to unfair or discriminatory decisions.
 - **Data Security:** Risk of data breaches and unauthorized access.
 - **Transparency:** Lack of clarity in how data is collected and used.
-

29. How BI Supports Competitive Advantage:

BI enables faster, data-driven decision-making, identifies market trends and customer needs ahead of competitors, optimizes internal operations, and reduces waste—leading to better performance and innovation.

30. Challenges Organizations Face When Implementing BI:

- **Data Quality and Integration:** Ensuring clean, unified data from multiple sources.
- **Cost and Complexity:** High initial investment and complexity in setup.
- **User Adoption:** Resistance from staff due to lack of training or fear of change.
- **Scalability:** Difficulty scaling BI solutions as the business grows.
- **Security and Compliance:** Managing data privacy and regulatory requirements.